



STEFANO RIVA

PhD, Research Scientist at Autodesk Research

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Research Scientist specializing in Physics-Informed AI/ML for engineering and design. Proven track record in developing reduced-order models and data-driven state estimation. Expertise in bridging traditional numerical solvers with machine learning to accelerate high-fidelity simulations.

EDUCATION

Bachelor's Degree in Energy Engineering

Politecnico di Milano

Milan, Ita

Oct. 2016 – Sep. 2019

Master's Degree *cum Laude* in Nuclear Engineering

Politecnico di Milano

Milan, Ita

Sep. 2019 – Dec. 2021

- Thesis Title: "Reduced basis methods for data assimilation in real thermal hydraulics systems"
- Supervisors: Prof. Antonio Cammi, Prof. Stefano Lorenzi, Dr. Carolina Introini

PhD Degree *cum Laude* in Energy and Nuclear Science and Technology

Politecnico di Milano

Milan, Ita

May 2022 - Oct. 2025

- Thesis title: "Advanced Data-Driven Techniques for State Estimation in Nuclear Reactors"
- Supervisors: Prof. Antonio Cammi, Dr. Carolina Introini, Prof. J. Nathan Kutz

Qualification to practice Engineering Profession

Milan, Ita

March 2024

RESEARCH AND TEACHING EXPERIENCE

Teaching Assistant

Fission Reactor Physics 1 course of prof. Antonio Cammi at Polimi

A.Y. 2022/2023 - 2023/2024, II Semester

Milan, Italy

VISIT Intern

Electrical and Computer Engineering Department, University of Washington

Oct. 2024 – March 2025

Seattle, USA

- Supervisor: Prof. J. Nathan Kutz
- Activities: SHallow REcurrent Decoder (SHRED) networks for State Estimation in Nuclear Reactors

Postdoctoral Researcher

Department of Energy, Politecnico di Milano

Aug. 2025 – March 2026

Milan, Italy

- Activities: development, verification and validation of Reduced-Order Models for two-phase flow simulations in the oil & gas sector, in collaboration with Eni Spa
- Supervisors: Prof. Antonio Cammi and Prof. Fabio Inzoli

Research Scientist – Physics-Informed AI

Autodesk Research, π -Labs

April 2026 – Present

London, UK

- Bridging data-driven AI approaches with traditional physics-based models for advanced simulation and design applications.

PROFESSIONAL AFFILIATIONS

Member of Associazione Italiana Nucleare (AIN)

Young Generation

Italy

2025 – Present

Member of American Nuclear Society (ANS)

Mathematics & Computation Division, Young Generation

USA

2025 – Present

LANGUAGE SKILLS

Native Italian speaker. Fluent in English (TOEIC Certification, February 2019, 930/990).

COMPUTATIONAL SKILLS

Operating system: MacOS, Linux, Windows
Languages: Python (advanced), C++ (intermediate), VisualBasic (intermediate), R (basic)
Numerical Computing Environments: MATLAB, Simulink
Finite Element Solvers: Dolfinx (Python Package), Freefem++ (basic)
Computational Fluid Dynamics: OpenFOAM, Ansys FLUENT (basic)
Machine Learning packages: scikit-learn, PyTorch
Developer Tools: Git, Visual Studio Code, Jupyter
Other: Microsoft Office, iMovie

RESEARCH INTERESTS

Scientific Machine Learning techniques for Fluid Dynamics and Nuclear Reactors
Multi-physics Modelling of Nuclear Reactor
Reduced Order Models and Data Assimilation
Computational Fluid Dynamics
Numerical Methods for Engineering

AWARDS

Best Master Thesis in Engineering <i>Cultural Association CISE2007</i>	27 September 2022
Best Paper Award <i>ICAPP-2023 Conference</i>	26 April 2023
Best Student Paper Award <i>NUTHOS-14 Conference</i>	28 August 2024
Young Author Award (co-author) <i>NENE2024 Conference</i>	12 September 2024
ENEN PhD Prize <i>European Nuclear Education Network (ENEN)</i>	15 May 2025
Young Professional Award <i>NURETH-21 Conference</i>	02 September 2025
Best Research Contribution Award (co-author) <i>SCOPE2 Conference</i>	05 November 2025
2nd Prize at Italian PhD Nuclear Talent Award <i>Associazione Italiana Nucleare (AIN, Italian Nuclear Association)</i>	12 March 2026

SELECTED GITHUB REPOSITORIES

pyISFenix <i>PYthon framework for the Incompressible Schrödinger Flow using FENICsX</i>	January 2024 Github , Documentation
pyforce <i>Python Framework for data-driven model Order Reduction of multi-physiCs problEms</i>	March 2024 Github , Documentation
NuSHRED <i>Shallow Recurrent Decoder for Nuclear Reactors applications</i>	September 2024 Github
PySHRED <i>Package for Shallow Recurrent Decoding</i>	June 2025 Github , Documentation
ctf4science <i>A modular and extensible platform designed for benchmarking methods on dynamic systems</i>	October 2025 Github , Documentation

VOLUNTEERING EXPERIENCE

Football Coach

A.S. Stella Azzurra '56

Sep. 2012 – June 2017

Cinisello Balsamo, Italy

Theatre Actor

Depramirao company

Apr. 2013 – Nov. 2014

Cinisello Balsamo, Italy

Adolescents' Educator

Oratorio San Luigi

Sep. 2014 – Sep. 2020, Sep. 2022 – Sep. 2024

Cinisello Balsamo, Italy

- [J1] C. Intorini, S. Cavalleri, S. Lorenzi, **Stefano Riva**, and A. Cammi, “Stabilization of Generalized Empirical Interpolation Method (GEIM) in presence of noise: A novel approach based on Tikhonov regularization,” *Computer Methods in Applied Mechanics and Engineering*, vol. 404, p. 115773, 2023. URL: <https://www.sciencedirect.com/science/article/pii/S0045782522007290>, doi:10.1016/j.cma.2022.115773.
- [J2] C. Intorini, **Stefano Riva**, S. Lorenzi, S. Cavalleri, and A. Cammi, “Non-intrusive system state reconstruction from indirect measurements: A novel approach based on hybrid data assimilation methods,” *Annals of Nuclear Energy*, vol. 182, p. 109538, 2023. URL: <https://www.sciencedirect.com/science/article/pii/S0306454922005680>, doi:10.1016/j.anucene.2022.109538.
- [J3] **Stefano Riva**, C. Intorini, S. Lorenzi, and A. Cammi, “Hybrid data assimilation methods, Part I: Numerical comparison between GEIM and PBDW,” *Annals of Nuclear Energy*, vol. 190, p. 109864, 2023. URL: <https://www.sciencedirect.com/science/article/pii/S0306454923001834>, doi:10.1016/j.anucene.2023.109864.
- [J4] **Stefano Riva**, C. Intorini, S. Lorenzi, and A. Cammi, “Hybrid Data Assimilation methods, Part II: Application to the DYNASTY experimental facility,” *Annals of Nuclear Energy*, vol. 190, p. 109863, 2023. URL: <https://www.sciencedirect.com/science/article/pii/S0306454923001822>, doi:10.1016/j.anucene.2023.109863.
- [J5] **Stefano Riva**, C. Intorini, and A. Cammi, “A finite element implementation of the incompressible Schrödinger flow method,” *Physics of Fluids*, vol. 36, p. 017138, 01 2024. doi:10.1063/5.0180356.
- [J6] A. Cammi, **Stefano Riva**, C. Intorini, L. Loi, and E. Padovani, “Data-driven model order reduction for sensor positioning and indirect reconstruction with noisy data: Application to a Circulating Fuel Reactor,” *Nuclear Engineering and Design*, vol. 421, p. 113105, 2024. URL: <https://www.sciencedirect.com/science/article/pii/S002954932400205X>, doi:10.1016/j.nucengdes.2024.113105.
- [J7] L. Loi, **Stefano Riva**, C. Intorini, F. Giacobbo, X. Wang, and A. Cammi, “OFELIA: An OpenMC-FEniCSx coupling for neutronic calculation with temperature feedback,” *Nuclear Engineering and Design*, vol. 428, p. 113480, 2024. URL: <https://www.sciencedirect.com/science/article/pii/S0029549324005806>, doi:10.1016/j.nucengdes.2024.113480.
- [J8] **Stefano Riva**, C. Intorini, and A. Cammi, “Multi-physics model bias correction with data-driven reduced order techniques: Application to nuclear case studies,” *Applied Mathematical Modelling*, vol. 135, pp. 243–268, 2024. URL: <https://www.sciencedirect.com/science/article/pii/S0307904X24003196>, doi:10.1016/j.apm.2024.06.040.
- [J9] M. Lo Verso, **Stefano Riva**, C. Intorini, E. Cervi, F. Giacobbo, L. Savoldi, M. Di Prinzio, M. Caramello, L. Barucca, and A. Cammi, “Application of a non-intrusive reduced order modeling approach to magnetohydrodynamics,” *Physics of Fluids*, vol. 36, p. 107167, 10 2024. doi:10.1063/5.0230708.
- [J10] **Stefano Riva**, C. Intorini, E. Zio, and A. Cammi, “Impact of malfunctioning sensors on data-driven reduced order modelling: Application to molten salt reactors,” *EPJ Web Conf.*, vol. 302, p. 17003, 2024. doi:10.1051/epjconf/202430217003.
- [J11] M. Lo Verso, **Stefano Riva**, C. Intorini, E. Cervi, L. Barucca, M. Caramello, M. Di Prinzio, F. Giacobbo, L. Savoldi, and A. Cammi, “Enhancing computational efficiency in nuclear fusion through reduced order modelling: Applications in magnetohydrodynamics,” *Fusion Engineering and Design*, vol. 216, p. 115080, 2025. URL: <https://www.sciencedirect.com/science/article/pii/S0920379625002777>, doi:10.1016/j.fusengdes.2025.115080.
- [J12] **Stefano Riva**, C. Intorini, A. Cammi, and J. N. Kutz, “Robust state estimation from partial out-core measurements with shallow recurrent decoder for nuclear reactors,” *Progress in Nuclear Energy*, vol. 189, p. 105928, 2025. URL: <https://www.sciencedirect.com/science/article/pii/S0149197025003269>, doi:10.1016/j.pnucene.2025.105928.

- [J13] M. A. Nasr, L. Loi, **Stefano Riva**, A. Zolfaghari, X. Wang, and A. Cammi, “Enhancing multi-physics modeling in new-generation nuclear reactors using machine learning: Implementing gaussian process regression for updating cross sections,” *Annals of Nuclear Energy*, vol. 224, p. 111720, 2025. URL: <https://www.sciencedirect.com/science/article/pii/S0306454925005377>, doi:10.1016/j.anucene.2025.111720.
- [J14] **Stefano Riva**, S. Deanesi, C. Introini, S. Lorenzi, and A. Cammi, “Real-time state estimation of neutron flux in molten salt fast reactors from out-core sparse measurements,” *Nuclear Science and Engineering*, vol. 0, no. 0, pp. 1–14, 2025. doi:10.1080/00295639.2025.2531477.
- [J15] **Stefano Riva**, C. Introini, E. Zio, and A. Cammi, “Data-driven reduced order modelling with malfunctioning sensors recovery applied to the molten salt reactor case,” *EPJ Nuclear Sci. Technol.*, vol. 11, p. 55, 2025. doi:10.1051/epjn/2025054.
- [J16] **Stefano Riva**, A. Missaglia, C. Introini, I. C. Bang, and A. Cammi, “A Comparative Study of Parametric Dynamic Mode Decomposition Algorithms on Thermal-Hydraulics Case Studies,” *Nuclear Technology*, vol. 0, no. 0, pp. 1–25, 2026. doi:10.1080/00295450.2025.2575701.
- [J17] **Stefano Riva**, C. Introini, and A. Cammi, “pyforce: Python framework for data-driven model order reduction of multi-physics problems,” *Journal of Open Source Software*, vol. 11, no. 117, p. 6950, 2026. doi:10.21105/joss.06950.
- [J18] M. Lo Verso, **Stefano Riva**, C. Introini, E. Cervi, M. Di Prinzio, M. Caramello, F. Giacobbo, L. Savoldi, and A. Cammi, “Parametric Dynamic Mode Decomposition for Nuclear Fusion: Applications to Magnetohydrodynamics,” *IEEE Transactions on Plasma Science*, pp. 1–6, 2026. doi:10.1109/TPS.2026.3682545.
- [J19] M. Alrwashdeh, C. Introini, **Stefano Riva**, X. Wang, F. Foulon, K. S. Chaudri, S. Al Ameri, and A. Cammi, “Adaptive Parametric Reconstruction with the Generalised Empirical Interpolation Method on the DYNASTY Loop with Limited Instrumentation,” *Nuclear Engineering and Technology*, p. 104606, 2026. URL: <https://www.sciencedirect.com/science/article/pii/S1738573326004948>, doi:10.1016/j.net.2026.104606.
- [J20] **Stefano Riva**, C. Introini, J. N. Kutz, and A. Cammi, “Constrained Sensing and Reliable State Estimation with Shallow Recurrent Decoders on a TRIGA Mark II Reactor,” *Chemical Engineering Science*, p. 125037, 2026. URL: <https://www.sciencedirect.com/science/article/pii/S0009250926017525>, doi:10.1016/j.ces.2026.125037.

SELECTED PREPRINTS

- [P1] **Stefano Riva**, A. Missaglia, C. Introini, J. N. Kutz, and A. Cammi, “From models to experiments: Shallow recurrent decoder networks on the dynasty experimental facility,” 2026. submitted to Nuclear Engineering and Design. URL: <https://arxiv.org/abs/2503.08907>, arXiv:2503.08907.
- [P2] **Stefano Riva**, C. Introini, A. Cammi, D. Price, A. Yermakov, Y. Zhao, P. M. Wyder, J. Goldfeder, J. Williams, A. S. Rude, M. Tomasetto, J. Germany, J. Bakarji, G. Maierhofer, M. Cranmer, and J. N. Kutz, “CTF4Nuclear: Common Task Framework for Nuclear Fission and Fusion Models,” 2026. URL: <https://arxiv.org/abs/2605.15549>, arXiv:2605.15549.
- [P3] **Stefano Riva**, C. Introini, J. N. Kutz, and A. Cammi, “Multi-Fidelity Learning with Shallow Recurrent Decoders for Multi-Physics Applications,” 2026. URL: <https://arxiv.org/abs/2606.05202>, arXiv:2606.05202.
- [P4] C. Scardino, **Stefano Riva**, C. Introini, M. L. Verso, E. Cervi, A. Cammi, and L. Savoldi, “Real-Time Monitoring of MHD Liquid Metal Flows with Shallow Recurrent Decoders,” 2026. URL: <https://arxiv.org/abs/2608.28366>, arXiv:2608.28366.

PROCEEDINGS OF INTERNATIONAL CONFERENCES

- [C1] **Stefano Riva**, A. Cammi, C. Introini, and S. Lorenzi, “Hybrid Data Assimilation Methods: Application to the DYNASTY Experimental Facility,” in *13th International Topical Meeting of Nuclear Reactor Thermal-Hydraulics, Operation and Safety (NUTHOS)*, (Taipei, Taiwan), pp. 505–518, September 2022.

- [C2] **Stefano Riva**, A. Cammi, and C. Introini, “Inviscid Fluid Simulation through *Incompressible Schrödinger Flow*: a Finite Element approach,” in *31st International Conference Nuclear Energy for New Europe (NENE2022)*, (Portoroz, Slovenia), September 2022.
- [C3] A. Cammi, **Stefano Riva**, C. Introini, L. Loi, and E. Padovani, “Indirect Field Reconstruction and Sensor Positioning in Circulating Fuel Reactors using Data-Driven Model Order Reduction,” in *2023 International Congress on Advances in Nuclear Power Plants*, (Gyeongju, Korea), April 2023.
- [C4] **Stefano Riva**, C. Introini, and A. Cammi, “Multi-Physics Model Correction with Data-Driven Reduced Order Modelling,” in *32nd International Conference Nuclear Energy for New Europe (NENE2023)*, (Portoroz, Slovenia), September 2023.
- [C5] **Stefano Riva**, L. Loi, C. Introini, A. Cammi, and X. Wang, “FEniCSx-OpenMC Coupling for Neutronic Calculation with Temperature Feedback,” in *32nd International Conference Nuclear Energy for New Europe (NENE2023)*, (Portoroz, Slovenia), September 2023.
- [C6] L. Loi, **Stefano Riva**, C. Introini, A. Cammi, and E. Padovani, “OpenMC Analysis of TRIGA Mark II Reactor Void and Temperature Reactivity Coefficients,” in *32nd International Conference Nuclear Energy for New Europe (NENE2023)*, (Portoroz, Slovenia), September 2023.
- [C7] **Stefano Riva**, S. Deanesi, C. Introini, S. Lorenzi, and A. Cammi, “Neutron flux reconstruction from out-core sparse measurements using data-driven reduced order modelling,” in *Proceedings of the International Conference on Physics of Reactors, PHYSOR 2024*, p. 1632 – 1641, 2024. doi:10.13182/PHYSOR24-43444.
- [C8] L. Loi, **Stefano Riva**, C. Introini, E. Padovani, F. Giacobbo, and A. Cammi, “An alternative approach for group constants regression based on supervised learning techniques,” in *Proceedings of the International Conference on Physics of Reactors, PHYSOR 2024*, p. 1674 – 1683, 2024. doi:10.13182/PHYSOR24-43521.
- [C9] **Stefano Riva**, A. Missaglia, C. Introini, I. C. Bang, and A. Cammi, “A Novel Approach for Parametric Dynamic Mode Decomposition: Application to the DYNASTY Experimental Facility,” in *The 14th International Topical Meeting on Nuclear Reactor Thermal-Hydraulics, Operation, and Safety (NUTHOS-14)*, (Vancouver, Canada), August 2024. doi:doi.org/10.13182/NUTHOS14-231.
- [C10] C. G. De Lurion De L’Égouthail, L. Loi, **Stefano Riva**, C. Introini, and A. Cammi, “Shadowing Effect Correction for the Pavia TRIGA Reactor Using Monte Carlo Data and Reduced Order Modelling Techniques,” in *The 33rd International Conference Nuclear Energy for New Europe (NENE2024)*, (Portoroz, Slovenia), September 2024.
- [C11] R. Boccelli, L. Loi, **Stefano Riva**, C. Introini, S. Lorenzi, and A. Cammi, “Analysis of KRUSTY reactor behaviour with OFELIA environment,” in *The 33rd International Conference Nuclear Energy for New Europe (NENE2024)*, (Portoroz, Slovenia), September 2024.
- [C12] **Stefano Riva**, C. Introini, J. N. Kutz, and A. Cammi, “Shallow Recurrent Decoders for State Estimation in Parametric Accidental Scenarios of Circulating Fuel Nuclear Reactors,” in *21st International Topical Meeting on Nuclear Reactor Thermal Hydraulics (NURETH-21)*, (Busan, Korea), September 2025.
- [C13] **Stefano Riva**, A. Missaglia, C. Introini, J. N. Kutz, and A. Cammi, “Verification and Validation of Shallow Recurrent Decoders for State Estimation in the DYNASTY facility,” in *21st International Topical Meeting on Nuclear Reactor Thermal Hydraulics (NURETH-21)*, (Busan, Korea), September 2025.
- [C14] Y. Xu, C. Introini, **Stefano Riva**, X. Zeng, M. Peng, Y. Wang, G. Xia, and A. Cammi, “Parametric Reduced-Order Modeling of Once-Through Steam Generators using Proper Orthogonal Decomposition and LSTM Neural Networks,” in *21st International Topical Meeting on Nuclear Reactor Thermal Hydraulics (NURETH-21)*, (Busan, Korea), September 2025.
- [C15] M. Lo Verso, **Stefano Riva**, C. Introini, E. Cervi, F. Giacobbo, M. Di Prinzio, M. Caramello, L. Savoldi, and A. Cammi, “Parametric Model Discovery in Magneto-Hydrodynamics,” in *21st International Topical Meeting on Nuclear Reactor Thermal Hydraulics (NURETH-21)*, (Busan, Korea), September 2025.

- [C16] P. M. Wyder, J. A. Goldfeder, A. Yermakov, Y. Zhao, **Stefano Riva**, J. P. Williams, D. Zoro, A. S. Rude, M. Tomasetto, J. Germany, J. Bakarji, G. Maierhofer, M. Cranmer, and J. N. Kutz, “Common Task Framework For a Critical Evaluation of Scientific Machine Learning Algorithms,” in *The 39th Annual Conference on Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, November 2025. URL: <https://openreview.net/forum?id=pkFKjmUE3L>.
- [C17] A. Yermakov, Y. Zhao, M. Denolle, Y. Ni, P. M. Wyder, J. A. Goldfeder, **Stefano Riva**, J. P. Williams, D. Zoro, A. S. Rude, M. Tomasetto, J. Germany, J. Bakarji, G. Maierhofer, M. Cranmer, and J. N. Kutz, “The Seismic Wavefield Common Task Framework,” in *The Fourteenth International Conference on Learning Representations*, 2025. accepted. URL: <https://openreview.net/forum?id=u4N7Kl6gzE>.
- [C18] **Stefano Riva**, C. Introini, J. N. Kutz, and A. Cammi, “State Estimation during Different Accidental Scenarios in a Molten Salt Fast Reactor with Shallow Recurrent Decoders from Out-Core Temperature Measurements,” in *International Conference on the Physics of Reactors, PHYSOR 2026*, (Turin, Italy), April 2026. doi:10.5281/zenodo.20804138.
- [C19] **Stefano Riva**, C. Introini, J. N. Kutz, and A. Cammi, “Multi-Fidelity Learning with Shallow Recurrent Decoders for Reactor Physics,” in *International Conference on the Physics of Reactors, PHYSOR 2026*, (Turin, Italy), April 2026. doi:10.5281/zenodo.20804140.
- [C20] Y. Luo, C. Introini, M. A. Nasr, L. Loi, W. Xiang, **Stefano Riva**, and A. Cammi, “Verification and Validation of FVM-based Neutronic Solvers against TRIGA Mark II Reactor,” in *International Conference on the Physics of Reactors, PHYSOR 2026*, (Turin, Italy), April 2026. doi:10.5281/zenodo.20803584.
- [C21] Y. Luo, C. Introini, L. Loi, M. A. Nasr, W. Xiang, **Stefano Riva**, and A. Cammi, “Preliminary Research on Spatial and Energy-group MGXS Homogenization for the TRIGA Mark II reactor,” in *International Conference on the Physics of Reactors, PHYSOR 2026*, (Turin, Italy), April 2026. doi:10.5281/zenodo.20803757.
- [C22] S. Deanesi, D. Pizzocri, **Stefano Riva**, and S. Lorenzi, “A Perturbation Theory Based Analysis of Reactivity Feedbacks during a ULOHS Transient in Molten Salt Reactors,” in *International Conference on the Physics of Reactors, PHYSOR 2026*, (Turin, Italy), April 2026. doi:10.5281/zenodo.20803347.

WORKSHOPS AND OTHER CONFERENCES

- [CN1] **Stefano Riva**, C. Introini, L. Marocco, L. Savoldi, and A. Cammi, “Inclusion of the buoyancy forces in the Incompressible Schrödinger Flow algorithm to simulate inviscid fluids,” in *41st UIT International Heat Transfer Conference*, (Naples, Italy), June 2024.
- [CN2] **Stefano Riva**, C. Introini, X. Wang, and A. Cammi, “Advection-Diffusion of Scalars with the Incompressible Schrödinger Flow,” in *The 14th International Topical Meeting on Nuclear Reactor Thermal-Hydraulics, Operation, and Safety (NUTHOS-14)*, (Vancouver, Canada), August 2024.
- [CN3] C. Introini, **Stefano Riva**, L. Loi, X. Wang, and A. Cammi, “Learning the dynamics of un-observable fields from out-core measurements of simple fields using Supervised Learning,” in *The 11th International Symposium on Symbiotic Nuclear Power: AI Application in Nuclear Systems (ISSNP2024)*, (Harbin, Heilongjiang, China), September 2024.
- [CN4] C. Introini, **Stefano Riva**, L. Loi, X. Wang, and A. Cammi, “State estimation in the DYNASTY experimental facility using Supervised Learning,” in *The 11th International Symposium on Symbiotic Nuclear Power: AI Application in Nuclear Systems (ISSNP2024)*, (Harbin, Heilongjiang, China), September 2024.
- [CN5] P. M. Wyder, J. A. Goldfeder, A. Yermakov, Y. Zhao, **Stefano Riva**, J. P. Williams, D. Zoro, A. S. Rude, M. Tomasetto, J. Germany, J. Bakarji, G. Maierhofer, M. Cranmer, and J. N. Kutz, “Common Task Framework For a Critical Evaluation of Scientific Machine Learning Algorithms,” in *Championing Open-source DEvelopment in ML Workshop @ ICML25*, 2025. URL: <https://openreview.net/forum?id=licgHdwg5y>.

- [CN6] C. Introini, C. Stecchetti, **Stefano Riva**, A. Cammi, F. Foulon, and M. Alrwashdeh, “Dynamics of the Molten Salt Fast Reactor Through Parametric Dynamic Mode Decomposition,” in *The 2nd Saudi International Conference on Nuclear Power Engineering (SCOPE)*, (Dhahran, Saudi Arabia), November 2025.
- [CN7] **Stefano Riva**, C. Introini, A. Cammi, X. Wang, F. Foulon, and M. Alrwashdeh, “Development of a Buoyancy-Driven Incompressible Schrodinger Flow Algorithm for Inviscid Flows,” in *The 2nd Saudi International Conference on Nuclear Power Engineering (SCOPE)*, (Dhahran, Saudi Arabia), November 2025.
- [CN8] A. Cammi, C. Introini, **Stefano Riva**, X. Wang, F. Foulon, and M. Alrwashdeh, “Robust Model Identification for Pressurized Water Reactor Systems using Sparse Identification of Nonlinear Dynamics,” in *The 2nd Saudi International Conference on Nuclear Power Engineering (SCOPE)*, (Dhahran, Saudi Arabia), November 2025.
- [CN9] C. Introini, **Stefano Riva**, A. Cammi, X. Wang, F. Foulon, and M. Alrwashdeh, “The Data-Driven Reduced Order Modelling Framework: Application on the DYNASTY Experimental Facility,” in *The 2nd Saudi International Conference on Nuclear Power Engineering (SCOPE)*, (Dhahran, Saudi Arabia), November 2025.
- [CN10] **Stefano Riva**, C. Introini, M. Lo Verso, J. N. Kutz, and A. Cammi, “Robust Monitoring of Nuclear Reactors from Sparse Sensors using Shallow Recurrent Decoders,” in *The 17th World Congress on Computational Mechanics & 10th European Congress on Computational Methods in Applied Sciences and Engineering*, July 2026.
- [CN11] M. Lo Verso, C. Introini, **Stefano Riva**, J. N. Kutz, and A. Cammi, “Application of Shallow Recurrent Decoder Networks for parametric state reconstruction of magnetohydrodynamic flows in liquid metal blankets of fusion reactors,” in *The 17th World Congress on Computational Mechanics & 10th European Congress on Computational Methods in Applied Sciences and Engineering*, July 2026.
- [CN12] Y. Luo, C. Introini, **Stefano Riva**, X. Wang, and A. Cammi, “Reduced-Order Modeling for Sensor Placement in the TRIGA Mark II Research Reactor under Region Accessibility Constraints,” in *10th International Conference on Reliability Engineering (ICRE2026)*, (Hangzhou, China), July 2026.
- [CN13] P. Conti*, E. Özalp*, **Stefano Riva***, and J. N. Kutz, “Data-driven reduced-order modeling with latent geometry encoding for inverse design,” in *Model Reduction and Surrogate Modeling (MORe 2026)*, 2026.
*These authors contributed equally to this work and are listed in alphabetical order.

CO-SUPERVISION OF MASTER’S THESES

- [TC1] C. de Lurion de l’Egouthail, “Shadowing Effect Correction for the Pavia TRIGA Reactor Using Monte Carlo Data and Reduced Modelling Techniques,” Master’s thesis, Politecnico di Milano, October 2024. Supervisor: Prof. A. Cammi, Co-Supervisors: L. Loi, S. Riva.
- [TC2] C. Stecchetti, “Model Discovery for Fluid and Reactor Dynamics with Parametric Dynamic Mode Decomposition,” Master’s thesis, Politecnico di Milano, July 2025. Supervisor: Prof. A. Cammi, Co-Supervisors: Dr. C. Introini, S. Riva.
- [TC3] F. Bowcott, “Predicting spatial-temporal systems from sparse and movable observation points using deterministic machine learning methods,” Master’s thesis, Imperial College London, August 2025. Supervisor: Prof. S. Cheng, Prof. R. Arcucci, Co-Supervisors: S. Riva, Dr. C. Introini.
- [TC4] C. Scardino, “State Estimation in Magnetohydrodynamics: from interpolation to recurrent networks,” Master’s thesis, Politecnico di Milano, March 2026. Supervisor: Prof. A. Cammi, Co-Supervisors: Dr. S. Riva, Dr. C. Introini.
- [TC5] D. Valisa, “Data-Driven Signal Reconstruction for Fault Tolerance in Molten Salt Reactors: an Application of Guassian Processes and Reduced Order Modeling,” Master’s thesis, Politecnico di Milano, March 2026. Supervisor: Prof. A. Cammi, Co-Supervisors: Dr. C. Introini, Dr. S. Riva.

- [TC6] T. Pirola, “Initial development and verification of a Digital Twin for Molten Salt Fast Reactor under accidental conditions,” Master’s thesis, Politecnico di Milano, July 2026. Supervisor: Prof. A. Cammi, Co-Supervisors: Dr. C. Introini, Dr. S. Riva.

PEER-REVIEW ACTIVITY

Reviewer of manuscripts for international peer-reviewed journals in nuclear engineering, fluid dynamics, computational methods, and scientific machine learning (*Nuclear Engineering and Design*, *Physics of Fluids*, *Applied Mathematical Modelling*, *Computer Physics Communications*, *Journal of Computational Physics*, *Annals of Nuclear Energy*, *Applied Energy*, and *Progress in Nuclear Energy*).